SLIDE 1/8 BAR-RAISER INTERVIEW CALIBRATION

Tier-1 AI System Design: 5 Critical Traps

The exact system design questions where 90% of candidates give toy answers — and how to answer like a Principal AI Architect.

The Trap

Giving standard YouTube tutorial answers to Staff-level questions.

The Bar

Demonstrating deep knowledge of failure modes, latency budgets, and cost modeling.

SLIDE 2/8 QUESTION 1: LATENCY UNDER LOAD

Q1: 'How do you handle 5,000 req/sec with a 2-second LLM?'

Testing your understanding of async architectures.

Toy Candidate Answer

'I will scale out my FastAPI pods on Kubernetes and call OpenAI.' (Fails immediately due to rate limits and massive bills).

Bar-Raiser Answer

Implement Semantic Caching (Redis) to absorb 35% repeat queries + Kafka queue for asynchronous jobs + vLLM continuous batching on private GPU cluster with speculative decoding.

SLIDE 3/8 QUESTION 2: HALLUCINATION MITIGATION

Q2: 'How do you prevent catastrophic hallucinations in production?'

Testing your understanding of guardrails and verification loops.

Toy Candidate Answer

'I added 'Do not make up facts' in the system prompt.' (Zero engineering value).

Bar-Raiser Answer

Multi-layered guardrail: Structural Pydantic JSON validation + Self-RAG reflection node (checks if answer is supported by retrieved context) + Automated Ragas evaluation in CI/CD.

SLIDE 4/8 QUESTION 3: COST OPTIMIZATION

Q3: 'Our LLM bill is $80k/month. How do you cut it by 70%?'

Testing your model routing and tiering strategy.

Toy Candidate Answer

'Switch from GPT-4 to GPT-4o-mini everywhere.' (Degrades quality on complex tasks).

Bar-Raiser Answer

Tiered Model Router: Use a lightweight 8B local model for classification and intent parsing. Route 70% of simple tasks to fast 8B models, and only route complex multi-step reasoning to frontier models + Aggressive Cohere Reranking to cut prompt token size by 60%.

SLIDE 5/8 QUESTION 4: CONTEXT DEGRADATION

Q4: 'Why is our 128k context window missing key facts?'

Testing your knowledge of the 'Lost in the Middle' phenomenon.

Toy Candidate Answer

'Just increase the context window to 1M tokens.' (Increases latency 5x and worsens needle retrieval).

Bar-Raiser Answer

Address 'Lost in the Middle' bias: Place the most critical instructions at the very beginning and very end of the prompt + Apply contextual chunk compression + Structure retrieval into small, dense summaries.

SLIDE 6/8 QUESTION 5: MULTI-AGENT FAILURE ISOLATION

Q5: 'How do you prevent agent swarms from cascading failures?'

Testing your distributed systems resilience.

Toy Candidate Answer

'Restart the script if it crashes.'

Bar-Raiser Answer

Implement LangGraph with PostgreSQL checkpointing: every node execution is atomic. If Agent 3 fails, the state pauses without losing prior steps, triggers a fallback model, and requests human escalation after 2 retries.

SLIDE 7/8 THE BAR-RAISER SCORECARD

What Tier-1 Interviewers Grade You On

The 4 dimensions of an L6/Staff AI System Design interview.

1. Tradeoff Articulation

Can you explain why you chose BM25+Dense over pure Dense vectors?

2. Cost & Latency Modeling

Can you calculate token throughput and GPU VRAM requirements on the fly?

3. Failure Mode Thinking

Do you design for rate limits, API timeouts, and prompt injections by default?

4. Observability & Evals

Can you design a production monitoring and CI/CD evaluation harness?

SLIDE 8/8 PRACTICE WITH REAL VIVA DEFENSE

Defend Your System Design with AI Voice Viva

The only platform with automated 1:1 voice defense calibrated to Google Bar-Raisers.

Voice AI Bar-Raiser

Our Voice AI grills your architecture, tests your edge cases, and gives instant feedback.

6 Production Repos

Build the exact systems interviewers ask about (RAG, Agents, vLLM Serving).